AIML 501 // VISUAL FRAMEWORK

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MACHINE LEARNING ALGORITHMS, MAPPED.

Each algorithm is classified by learning type, by application domain, by real-world use cases, and by how it works. Lime badges are supervised, outlined badges are unsupervised. Colored tags mark the domain.

01 WHERE THESE ALGORITHMS SIT

NESTED FIELDS: GENERAL TO SPECIFIC

ARTIFICIAL INTELLIGENCE

Any system that performs tasks needing human-like intelligence

MACHINE LEARNING

Algorithms that learn patterns from data instead of hand-written rules. All 12 below live here.

DEEP LEARNING

ML using multi-layer neural networks: CNNs, Transformers, Autoencoders, GANs

TWO LEARNING TYPES: THE KEY DIFFERENCE

SUPERVISED LEARNING

Trains on **labeled** data: every example comes with the right answer. Goal: predict a label or number for new data. Tasks: classification, regression.

UNSUPERVISED LEARNING

Trains on **unlabeled** data: no answers given. Goal: find hidden structure. Tasks: clustering, dimensionality reduction, generation.

DOMAIN LEGEND

TABULAR VISION NLP GEN AI

02 SUPERVISED ALGORITHMS

LABELED DATA

Linear Regression

SUPERVISED · REGRESSION

TABULAR

How it works: Fits a straight line (or plane) through the data that minimizes the squared distance to each point, then uses that line to predict a continuous number.

Used for: Predicting house prices, sales forecasting, estimating demand from ad spend.

Logistic Regression

SUPERVISED · CLASSIFICATION

TABULAR NLP

How it works: Like linear regression, but passes the result through a sigmoid curve to output a probability between 0 and 1, then picks a class with a threshold.

Used for: Spam vs. not spam, loan default risk, customer churn, disease yes/no screening.

Decision Tree

SUPERVISED · BOTH

TABULAR

How it works: Repeatedly splits the data on the feature that best separates the outcomes (for example, "income > $50K?"), forming a flowchart of yes/no questions that ends in a prediction.

Used for: Credit approval rules, medical triage, customer segmentation explanations.

Random Forest

SUPERVISED · BOTH

TABULAR

How it works: Builds hundreds of decision trees, each on a random sample of rows and features, then lets them vote. Averaging many trees cancels out the errors of any single one.

Used for: Fraud detection, predicting equipment failure, ranking features that matter most.

Support Vector Machine

SUPERVISED · CLASSIFICATION

TABULAR NLP

How it works: Finds the boundary between classes that leaves the widest possible gap (margin) to the nearest points. A "kernel trick" lets it draw curved boundaries in higher dimensions.

Used for: Text categorization, handwriting recognition, bioinformatics classification.

K-Nearest Neighbors

SUPERVISED · BOTH

TABULAR VISION

How it works: No real training step. To predict a new point, it looks at the K closest labeled examples and takes a majority vote (classification) or average (regression).

Used for: Recommendation ("users like you"), simple image matching, anomaly flags.

Convolutional Neural Net

SUPERVISED · DEEP

VISION

How it works: Slides small filters across an image to detect edges, then textures, then shapes, layer by layer. Pooling shrinks the image so later layers see the big picture.

Used for: Face recognition, medical scan analysis, self-driving object detection (YOLO, ResNet).

Transformer (GPT, BERT)

SUPERVISED* · DEEP

NLP GEN AI

How it works: Uses "attention" to weigh how every word relates to every other word in a sequence, then predicts the next token. *Pretrained on raw text (self-supervised), fine-tuned with labels.

Used for: Chatbots, translation, summarization, code generation, sentiment analysis.

03 UNSUPERVISED ALGORITHMS

UNLABELED DATA

K-Means Clustering

UNSUPERVISED · CLUSTERING

TABULAR VISION

How it works: Picks K center points, assigns every record to its nearest center, moves each center to the middle of its group, and repeats until the groups stop changing.

Used for: Customer segmentation, image color compression, grouping similar documents.

Principal Component Analysis

UNSUPERVISED · REDUCTION

TABULAR VISION

How it works: Finds the directions in which the data varies the most and projects everything onto those few directions, so 100 columns can become 2 or 3 with most of the information kept.

Used for: Visualizing high-dimensional data, speeding up other models, noise reduction.

Autoencoder

UNSUPERVISED · DEEP

VISION GEN AI

How it works: A neural network squeezes the input into a small code, then tries to rebuild the original from it. The squeeze forces it to learn what matters; the rebuild can also generate new samples.

Used for: Anomaly detection, image denoising, feature learning, generating variations (VAEs).

Generative Adversarial Net

UNSUPERVISED · DEEP

VISION GEN AI

How it works: Two networks compete: a generator makes fake samples and a discriminator tries to spot them. Each round the generator gets better at fooling the judge until its output looks real.

Used for: Realistic image synthesis, photo upscaling, synthetic training data, deepfake detection research.

04 HOW THEY COMPARE

ALGORITHM	TYPE	OUTPUT	EXPLAINABLE?	COMPLEX PATTERNS?	DATA NEEDED
Linear Regression	Supervised	Number	Yes	No (linear only)	Small
Logistic Regression	Supervised	Class + probability	Yes	No (linear boundary)	Small
Decision Tree	Supervised	Class or number	Yes (flowchart)	Some, tends to overfit	Small to medium
Random Forest	Supervised	Class or number	Partly	Yes	Medium
SVM	Supervised	Class	Partly	Yes (with kernels)	Small to medium
KNN	Supervised	Class or number	Yes	Some, slow on big data	Small to medium
CNN	Supervised (DL)	Class, boxes, masks	No	Yes (images)	Large
Transformer	Self-sup. + Supervised (DL)	Text, tokens	No	Yes (sequences)	Very large
K-Means	Unsupervised	Cluster labels	Yes	No (round clusters)	Any
PCA	Unsupervised	Fewer features	Partly	No (linear only)	Any
Autoencoder	Unsupervised (DL)	Code / rebuilt data	No	Yes	Large
GAN	Unsupervised (DL)	New samples	No	Yes	Large

Classic vs. deep: Regression, trees, SVM, KNN, K-Means and PCA work well on small tabular datasets and are easier to interpret. CNNs, Transformers, Autoencoders and GANs need lots of data and compute but handle images and language.

Same family, different jobs: Linear and Logistic Regression share the same math; one outputs a number, the other a probability. A Random Forest is just many Decision Trees voting.

Labels decide the type: If you have the answers, use supervised. If you only have raw data and want to discover groups, compress it, or generate new examples, use unsupervised.